Three Coins Again

[12 pts]

Q1. First coin

The coin falls from rest through height h, so just before impact its speed is

$$v = \sqrt{2gh} = \sqrt{2 \cdot 10 \cdot 0.45} = 3 \text{ m/s}.$$

Perfectly inelastic collision (gravity's impulse neglected):

$$mv = (M+m)V_1 \implies V_1 = \frac{mv}{M+m} = \frac{0.02 \cdot 3}{0.12} = \boxed{0.5 \text{ m/s (downward)}}.$$

The equilibrium compression changes. With the tray alone it is $y_{\text{eq1}} = Mg/k = 1.25\,\text{cm}$; with the coin it becomes $y_{\text{eq2}} = (M+m)g/k = 1.5\,\text{cm}$. At the moment of impact the tray sits at y_{eq1}, i.e. a displacement $\delta = y_{\text{eq1}} - y_{\text{eq2}} = -0.25\,\text{cm}$ from the new equilibrium, moving at V_1 downward. With angular frequency $\omega_1 = \sqrt{k/(M+m)}$,

$$A_1 = \sqrt{\delta^2 + \left(\frac{V_1}{\omega_1}\right)^2} = \sqrt{(0.0025)^2 + \left(\frac{0.5}{25.82}\right)^2} = \boxed{1.95 \text{ cm}}.$$

Q2. Second coin

At the first return to the tray's initial position the system passes through the same displacement δ, so by energy conservation its speed equals the speed it had just after the first collision:

$$\boxed{v_2 = 0.5 \text{ m/s}}.$$

Q3. Third coin

Coin 2 falls at 3 m/s downward, while the system is moving 0.5 m/s *upward*. Their momenta cancel:

$$m(3) + (M+m)(-0.5) = 0.02 \cdot 3 - 0.12 \cdot 0.5 = 0,$$

so the system is momentarily at rest after the second collision. It then oscillates about the new equilibrium $y_{\text{eq3}} = (M+2m)g/k = 1.75\,\text{cm}$, starting at rest at $y_{\text{eq1}} = 1.25\,\text{cm}$, which is a turning point ($\delta = -0.5\,\text{cm}$). At its first return to the initial position (one full oscillation later) the system is again at rest.

Thus coin 3 strikes a stationary system at 3 m/s:

$$mv = (M+3m)V_3 \implies V_3 = \frac{0.02 \cdot 3}{0.16} = \boxed{0.375 \text{ m/s (downward)}}.$$

Q4. Separation?

After the third collision the total mass is $M + 3m = 0.16\,\text{kg}$, with equilibrium compression $y_{\text{eq4}} = (M+3m)g/k = 2.0\,\text{cm}$ and angular frequency $\omega_3 = \sqrt{k/(M+3m)}$. At impact the system is at $y_{\text{eq1}} = 1.25\,\text{cm}$ moving at V_3, so the amplitude is

$$A_3 = \sqrt{(1.25 - 2.0)^2 + \left(\frac{V_3}{\omega_3}\right)^2} = \sqrt{(0.0075)^2 + \left(\frac{0.375}{22.36}\right)^2} = 1.84 \text{ cm}.$$

The coin on top separates from the stack below only if the stack's downward acceleration exceeds g (a free coin can accelerate downward no faster than g). The maximum downward acceleration occurs at the top of the oscillation:

$$a_{\mathrm{max}} = \omega_3^2 A_3 = 9.19\ \mathrm{m/s^2} < g = 10\ \mathrm{m/s^2}.$$

Since $a_{\mathrm{max}} < g$, the contact force never drops to zero, so

$$\boxed{\text{coin 3 does not separate.}}$$